

Natural Language Processing

NLP enables computers to understand, interpret, and generate human language using machine learning techniques.

Tokenization breaks text into words or subwords. Modern tokenizers like BPE and WordPiece handle out-of-vocabulary words.

Transformer models like BERT, GPT, and T5 have revolutionized NLP with attention mechanisms and transfer learning.

Named entity recognition identifies persons, organizations, locations, and other entities in unstructured text.

NLP Applications

Sentiment analysis determines the emotional tone of text, widely used for social media monitoring and customer feedback.

Machine translation systems like Google Translate use sequence-to-sequence models with attention mechanisms.

Text summarization generates concise summaries from longer documents using extractive or abstractive approaches.

Question answering systems extract answers from context paragraphs, powering virtual assistants and chatbots.